Comparing Structural and Reduced-Form Credit Risk Models for Default Risk Measurement: Merton, VK, and Hull-White

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Background and Motivation

Credit risk modeling is central to fixed-income markets, credit derivatives pricing, and risk management. Broadly, credit risk models can be divided into two categories: structural models, which link default to firm asset dynamics (e.g., Merton and VK/KMV-style models), and reduced-form models, which model default as an intensity (hazard-rate) process (e.g., Hull-White style intensity models).

This project compares the empirical performance of three representative approaches:

- Merton structural model (baseline)
- Vasicek-Kealhofer (VK/KMV-style) structural model
- Hull-White reduced-form default intensity model

The comparison focuses on two dimensions:

- Default discrimination ability (whether the model can separate firms that default from those that do not)
- CDS spread prediction / relative value performance (how well model-implied CDS spreads match observed market CDS spreads)

This topic is highly relevant to fixed-income and credit markets because it connects equity information, balance-sheet fundamentals, and credit-derivative pricing. It also combines modeling, coding, and empirical validation.

Research Questions

This project seeks to answer the following questions:

1. Default Discrimination
    - Do VK and HW models provide better default discrimination than the baseline Merton model?
    - How large is the performance gap as measured by ROC/AUC and Accuracy Ratio (AR)?
2. CDS Spread Prediction
    - Which model generates CDS-implied spreads that are closest to observed 5Y CDS spreads?
    - Does model performance vary across issuers with different liquidity or leverage characteristics?
3. Modeling Trade-Offs
    - What are the practical strengths and limitations of structural vs. reduced-form models in a fixed-income / credit-risk context?

Data

The project design uses a multi-source panel combining equity, balance-sheet, risk-free-rate, and (optionally) CDS data, with a target sample of U.S. listed issuers (approximately 50–100 firms, including defaulted firms if possible).

For this course project, we will use a feasibility-first data strategy:

- Equity prices / returns / market cap proxy (Yahoo Finance)

- Balance-sheet data (public filings or Compustat, if available)
- Risk-free rates (FRED Treasury / SOFR / OIS proxy)
- CDS spreads (optional extension, if accessible)

This structure keeps the project executable even without full access to proprietary CDS databases. If CDS data is unavailable, the main contribution will remain the default-risk comparison across the three models.

Expected Deliverables

By the end of this project, we expect to produce a clean firm-level dataset (equity, balance-sheet, and risk-free-rate data, with CDS data as an optional extension) and a Python implementation of three model components: a Merton solver for nonlinear asset-value/asset-volatility estimation, a VK DD/PD module, and a Hull-White reduced-form hazard/survival module.

The project will generate model comparison outputs, including firm-level risk tables (DD/PD/hazard-based measures) and discrimination results using CAP / ROC / AUC / AR. If CDS data is available, we will also produce an optional CDS pricing comparison using model-implied versus observed 5Y CDS spreads. The final deliverable will be a short written report summarizing the methodology, empirical findings, limitations, and possible extensions.

This project is well suited for a fixed-income / credit-risk course because it connects firm fundamentals and equity market signals, hazard-rate and survival-probability modeling, and (optionally) credit-derivative pricing logic. More broadly, it provides a practical comparison of two major credit-risk modeling paradigms used in fixed-income markets and risk management.

References

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